

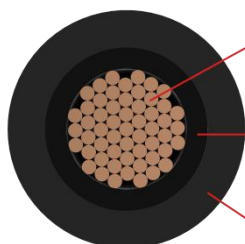


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HELUKABEL Solar Cable H1Z2Z2-K 1500V DC

(2.5 mm²-35 mm²)

Construction



Stranded tinned copper conductor, cl5, according to IEC 60228

Insulation: XLPO, electron-beam, cross-linked

Jacket: XLPO, electron-beam, cross-linked, UV and ozone resistance ,Bla

Dimensions (approx.)

Cross-section (mm ²)	Cu-Conductor Ø (mm)	Insulation Ø (mm)	Sheath Ø (mm)
2.5	1.95	3.40	5.10
4	2.45	3.85	5.50
6	3.00	4.45	6.10
10	4.00	5.50	7.20
16	5.05	6.50	8.10
25	6.80	8.70	10.80
35	8.10	10.00	12.40

Technical data

Construction (mm ²)	2.5	4	6	10	16	25	35
Max. Conductor resistance at 20°C(Ω/km)	8.21	5.09	3.39	1.95	1.24	0.795	0.565
Voltage drop 20°C loop (mV/m.A)	16.42	10.18	6.78	3.90	2.48	1.590	1.13
Voltage drop 90°C loop (mV/m.A)	20.94	12.98	8.65	4.97	3.16	2.03	1.44
Max. Cable weight (kg/km)	43	59	79	126	186	303	416

Note : The circuit is regarded as a loop and the current is calculated at 1Amps.

Specifications can be subject to change without notice



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Voltage rating according to EN 50618

Max. permitted voltage	U ₀	1.8KV (DC)
Nominal voltage	U ₀	1.5KV (DC)
	U ₀ /U	1/1 KV (AC, 50Hz)
Testing voltage	6.5 KV, AC, 50Hz 5 min	
Operating temperature	- 40 °C up to + 90 °C	
Min. bending radius	Fixed installation	> 6 x cable-Ø
	Occasionally moved	> 8 x cable-Ø
Maximal Maximal pull force	40 N / mm ² (1 x conductor x 40 N/mm ²)	

Max. Current rating (Ampere at 50 Hz) (according to IEC 60287) for AC&DC application

(Calculation is valid for thermal soil resistivity : 1.00 K.m/W and Load Factor : 1.00)

	in air (60°C) no sun	in air (60°C) no sun in plastic Pipe	in air (60°C) with sun	in air (60°C) with sun in plastic Pipe	(max 20°depth 1m) in plastic Pipe
Conductor (mm ²)	Temp. 90°C	Temp. 90°C	Temp. 90°C	Temp. 90°C	Temp. 90°C
2.5	42	38	40	36	53
4	55	49	52	45	68
6	70	64	66	59	86
10	97	86	93	80	117
16	129	115	122	107	152
25	170	147	160	136	193
35	214	183	201	169	238



	in air (60°C) no sun	in air (60°C) with sun	in Plastic-Pipe (max 20° depth 1m)
Conductor (mm²)	Temp 90°C	Temp. 90°C	Temp. 90°C
2.5	30	27	36
4	40	36	47
6	51	45	58
10	71	64	80
16	95	85	102
25	127	113	132
35	161	143	163

Conversion factors for different thermal soil resistivity

ρ (K. m/W)	0.50	0.70	1.00	1.50	2.00	2.50
Buried cables	1.25	1.13	1	0.85	0.76	0.68
Buried cables in pipe	1.08	1.05	1	0.93	0.87	0.83

Conversion factors for different ambient temperature and depth

Ambient Temperature (°C)	20	30	40	50	60	70	80	85
Factor	1.35	1.26	1.18	1.09	1.00	0.89	0.78	0.72

Conversion factors for parallel Installations

(Only valid for soil resistivity 1.0 K.m / W and and soil temperature 20 °C)

	Parallels cable systems (center distance 7 cm) 1 core							
Parallel laying in soil	1	2	3	4	5	6	8	10
Factor	1	0.87	0.77	0.73	0.70	0.68	0.65	0.63

	Parallels cable systems (center distance 15 cm) 3 core							
Parallel laying in soil	1	2	3	4	5	6	8	10
Factor	1	0.85	0.75	0.70	0.66	0.63	0.59	0.56

■ LAYING CONDITIONS CABLES

As laying depth, the mathematical distance to the cable axis – for triangular bunched laying the distance of the bundle axis to the earth surface with 70 cm is choised. With increased laying depth the load ratings will be mathematically reduced. Hereby the same temperature and the same thermal earth resistances are to be presumed. Normal operation conditions and indications for deviating operation conditions.

Normal operation conditions

Laid in Earth		Laid in Air		Indications
1 Multicore cable		1 Multicore cable		Conversions factors see the following tables
1 Single core cable in direct current-system		1 Single core cable in direct current system		as of collective laying conditions see the following tables
3 Single core cables in 3-phase system, side by side, with a space of 7 cm		3 Single core cables in 3-phase system, side by side, with a space of a cable ø		
3 Single core cables in 3-phase system, in bundle form ¹⁾		3 Single core cables in 3-phase system in bundle form ¹⁾		
Bedding in sand or earth shove and if necessary covering with bricks, cement plates or with flat to light curved thin covering of plastic		– Laid in open air, i.e. unhindered heat radiation will be ensured at: Distance of cable from wall, floor or ceiling ≥ 2 cm – For cables laying side by side: Space at least two times of the cable ø – For cables laying one above the other: Vertical space of the cable atleast two times of the cable ø cable length at least 30 cm – Consideration of thermal loss in cable, the increased air temperature of sufficient big and ventilated rooms – Protection against direct heatradiation of sunlight etc. – Air temperature 30°C	• Conversion factors for laying in earth: – covering hood with air cavat $y = 0,9$ - laid in conduit = 0,85 • Conversion factors for laying in air: – alternating ambient temperatures – as of collecting laying conditions – for laying in conduits – see tables and indications according to DIN VDE 0298	
Ambient conditions – Ground temperature at installation depth: 20°C – Soil-thermal resistivity of moist area: 1,0 K · m/W – Soil-thermal resistivity of dry area: 2,5 K · m/W				
Connecting and earthing of metal sheaths or screens on both ends				Connecting and earthing of metal sheaths or screens on both sides

¹⁾ in "bunched" or triangle touching arrangement

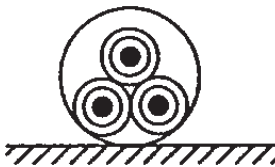


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Conversion factors for inside a cable ladder (According to VDE 0298-4)

	Number of cables					
	1	2	3	4	6	9
Parallel laying inside a trench touched	1	0.88	0.82	0.79	0.76	0.73
Factor	1	0.88	0.82	0.79	0.76	0.73

Conversion factors for bunched laying direct on the wall or floor , in electrical installation duct, on the wall or in the wall (acc. DIN VDE 0298-4:2013-06)

Bunched laying	Number of cables														
	1	2	3	4	5	6	7	8	9	10	12	14	16	18	20
  	1	0.80	0.70	0.65	0.60	0.57	0.54	0.52	0.50	0.48	0.45	0.43	0.41	0.39	0.38

Standard specifications / Material properties

Approval	EN 50618/IEC62930
Halogen free	IEC 62821-1:2015
Fire performance	IEC 60332-1-2:2004
Smoke emission	IEC 61034-2
UV resistant	HD 605

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